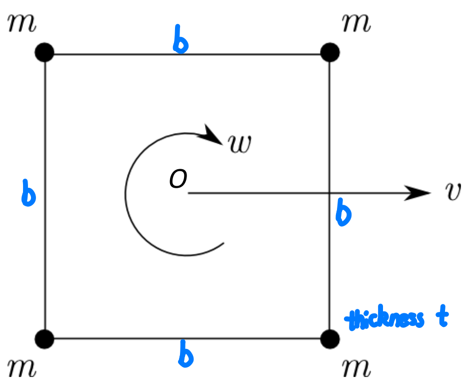
AERO40007 Progress Test 1

NAME	CID	GRADE (/100)

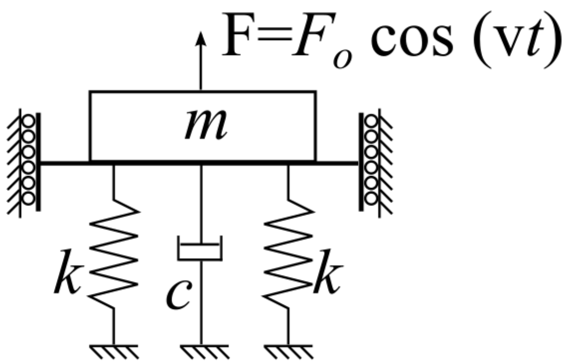
You can use the Datasheet available in Blackboard.

1. The figure shows a square body with side length b, thickness t and constant density ρ. Four point masses m are attached at the corners. The body has a constant horizontal velocity v and a constant angular velocity about the centre of mass ω.



   What is the total kinetic energy of the body?

2. Write the equation of motion. Calculate the natural frequency and damping ratio of the system below:



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60/100

1. $KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$\cancel{I = \Sigma mr^2 = \frac{1}{3}m[(\frac{1}{2}b)^2 + (\frac{1}{2}b)^2] + 4[2m(\frac{1}{2}b)^2 + (\frac{1}{2}b)^2 2m]}$

$\cancel{= \frac{1}{3}\rho b^2 t(\frac{1}{2}b^2) + 4(mb^2)}$

$\cancel{= \frac{1}{6}\rho b^4 t + 4mb^2}$

$I = \Sigma mr^2 = \frac{1}{12}b\cdot b^3\cdot\rho\cdot t + \underset{5}{\frac{1}{12}b\cdot b^3\cdot\rho\cdot t} + 4(m\cdot(\frac{b}{\sqrt{2}})^2) \underset{0}{=} \frac{1}{6}\rho b^4 t + 2mb^2$

$I = \Sigma mr^2 = \frac{1}{12}b\cdot b^3\cdot\rho\cdot t + 4[2m(\frac{1}{2}b)^2 + (\frac{1}{2}b)^2 2m]$

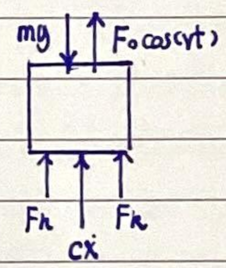
$= \frac{1}{12}\rho b^4 t + 4mb^2$

$\therefore KE = \frac{1}{2}(4m + \overset{10}{b^2 t\rho})v^2 + \frac{1}{2}(\overset{\frac{1}{6}\rho b^4 t}{\cancel{\frac{1}{12}\rho b^4 t}} + \overset{2mb^2}{\cancel{4mb^2}})\omega^2$

$= 2mv^2 + \overset{10}{\frac{1}{2}\rho b^2 t v^2} + \underset{12}{\cancel{\frac{1}{24}}}\rho b^4 t\omega^2 + \cancel{2}mb^2\omega^2$ 5

... 30

2. $m\ddot{x} + c\dot{x} + kx = -\overset{-mg}{\cancel{F_x}} = -F_0\cos(vt)\ \cancel{-mg}$ 10



$\therefore x(t) = A\cos(vt - \phi)$

$A = \frac{F_0}{\sqrt{(2k - mv^2)^2 + (cv)^2}}$

$\therefore x(t) = \frac{F_0}{\sqrt{(2k - mv^2)^2 + (cv)^2}}\cos(vt - \phi)$

$\omega_n = \sqrt{\frac{2k}{m}}$ 10

$\zeta = \frac{c}{2m\omega_n} = \frac{c}{2m}\sqrt{\frac{m}{2k}} = \cancel{\frac{\sqrt{2}c}{4}}\ \frac{\sqrt{2\cancel{m}}\,c}{4\sqrt{km}}$ 10

$\therefore ratio = \frac{T_{damped}}{T_{undamped}} = \frac{1}{\sqrt{1-\zeta^2}}$

$\because \sqrt{1-\zeta^2} = \sqrt{1 - \frac{2\cancel{m}c^2}{16km}} = \sqrt{1 - \frac{\cancel{m}c^2}{8km}}$

$\therefore ratio = (1 - \frac{\cancel{m}c^2}{8km})^{-1}$

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